

Assignment 1 - EfficientNet-B0 on UC Merced Land Use

Dataset Preparation

UC Merced Land Use dataset (HuggingFace: blanchon/UC_Merced): 2100 images, 21 classes, 100 per class. Stratified 70/15/15 split yields 1470 train / 315 val / 315 test images. Stratified splitting ensures each class is proportionally represented across all splits.

Training from Scratch

EfficientNet-B0 (~4M parameters), CrossEntropyLoss, AdamW lr=1e-4, 20 epochs, CPU training.

Final test accuracy: 55.87%. With only 70 training samples per class and random initialization, the model struggles to learn meaningful features. It shows gradual improvement but never fully converges. Basically too few images per class to build representations from scratch.

Training with Pretrained Weights

Same setup, ImageNet pretrained weights. Test accuracy: 96.51%, balanced accuracy: 96.51%.

The pretrained model already has strong feature representations from ImageNet. Fine-tuning on UC Merced is far more data-efficient, and the model converges within the first few epochs rather than slowly climbing for 20.

Comparison & Discussion

Gap: 96.51% vs 55.87% (~40 percentage points). Transfer learning makes an enormous difference with limited data (1470 training samples, 21 classes).

Hardest classes: dense/medium/sparse residential - from the air these all look like clusters of rooftops at different densities, which is genuinely ambiguous. Buildings vs parking lot were also confused since both appear as large flat rectangular structures from above.

Conclusion: with a small dataset, pretrained weights are basically necessary. Training from scratch requires far more data to reach comparable performance.